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Structural Dynamics Drive Mobility in Water-in-Salt Electrolytes

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Water-in-salt, electrolytes, lithium batteries, transport properties, multiscale dynamics, conductivity, QENS.

Abstract

Water-in-salt electrolytes (WISE) defy classical transport models by maintaining high ionic conductivity despite extreme viscosities. Here, we uncover the multiscale mechanisms behind this anomaly in superconcentrated aqueous Lithium bis(trifluoromethane)sulfonimide (LiTFSI) solutions. Using viscosity and conductivity measurements alongside Pulsed-Field Gradient Nuclear Magnetic Resonance (PFG-NMR) and Quasi Elastic Neutron Scattering (QENS), we reveal a pronounced decoupling between species mobility and bulk viscosity. Lithium species and protons retain high mobility, while TFSI⁻ anions are increasingly hindered. QENS identifies transient, water-rich nanodomains that enable rapid local diffusion and dynamic interdomain exchange. This work provides direct experimental evidence for hierarchical, decoupled transport in WISE, offering a new framework for designing high-performance electrolytes for energy storage.

1. Introduction

Water-in-salt electrolytes (WISE), first introduced by Suo et al.¹, are a turning point for aqueous battery technology. By dissolving lithium salts in water at high molalities (exceeding 5 mol·kg⁻¹), WISE enable electrochemical stability windows of up to 3 V, making them viable as electrolyte for high-voltage aqueous batteries^{2,3}. Beyond their electrochemical promise, WISE exhibit impressive transport properties, maintaining high ionic conductivity, even at extreme concentrations. For instance, in aqueous Lithium bis(trifluoromethane)sulfonimide (LiTFSI) solutions, conductivity remains above 10 mS·cm⁻¹ up to ~21 mol·kg⁻¹ despite viscosity above 40 mPa·s⁻¹⁴. Unlike dilute aqueous electrolytes, WISE systems exhibit microscopic structures and dynamical behaviours that deviate from ideal solution models, necessitating a detailed molecular understanding of both lithium ion and water dynamics.

Experimental and computational studies found out that increasing salt concentration beyond the solubility regime leads to a reorganization of the liquid “structure”. In LiTFSI WISE solution, most of water molecules are coordinated to Li⁺ ions, while the remaining water forms small, spatially confined clusters^{3,5,6}. Molecular dynamics (MD) simulations demonstrated that the extended hydrogen-bond network, which is generally attributed to bulk water, is largely disrupted, replaced by a heterogeneous nanostructure composed of lithium ion-rich domains⁷. The counter ions are then organized in a percolating anionic network forming apolar regions that coexist with polar Li-ion/water environments.

The consequences of this structural organization are visible especially in the transport properties since, the transport cannot anymore be seen as a vehicular diffusion model. Instead, the motion between ions and water molecules is correlated, leading to reduced water diffusion, irregular transference numbers, and deviations from Nernst–Einstein behaviour⁵. Furthermore, water dynamics become highly heterogeneous with some water molecules tightly bound to lithium solvation shells with hindered motion, while others undergo faster but spatially confined diffusion within transient water clusters. This behaviour has been attributed to the reduced activity of water and to the suppression of hydrogen evolution reaction (HER) at electrified interfaces.

From a theoretical standpoint, Bazant’s group showed that WISE electrolytes should be understood as “solvent-deficient ionic liquids” rather than concentrated aqueous solutions^{8,9}. Kornyshev and collaborators further developed this conceptual shift by drawing explicit parallels between WISE electrolytes and room-temperature ionic liquids. Their work emphasized the

importance of steric effects, ion–ion correlations, and short-range ordering in determining both bulk and interfacial properties^{10,11}. Thus, the reduced activity of water and the extended electrochemical stability window are natural consequences of constrained molecular motion³. Kornyshev's models of electric double layers in saturated electrolytes solutions predict layered interfacial structures in which water plays a secondary, highly constrained role, an idea supported by simulation and spectroscopic studies¹².

Unfortunately, experimental evidence about water dynamics (with relevant time and length scales) remains limited. Nuclear magnetic resonance (NMR) spectroscopy provides valuable information on long-time diffusion and exchange processes but are less sensitive to heterogeneous, picosecond-scale motion. Infrared and Raman spectroscopies probe vibrational signatures of hydrogen bonding but do not directly resolve translational dynamics. In this context, quasi-elastic neutron scattering (QENS) is the sole technique offering a unique probe of WISE. Indeed, hydrogen has a large incoherent scattering cross section making QENS a suitable sensitive technique to water motion, allowing direct measurement of translational diffusion, rotational relaxation, and confinement effects on time scales from picoseconds to nanoseconds and length scales from ångströms to nanometers.

Several studies combining QENS and molecular dynamics have demonstrated the ability of QENS to improve the understanding of water dynamics in highly concentrated electrolytes. These works revealed deviations from simple Fickian diffusion, stretched-exponential relaxation behaviour, and evidence for confined or sub-diffusive water motion, in quantitative agreement with atomistic simulations^{5,7,13-15}.

In this work, we address the question about transport dynamics in WISE solution using several advanced techniques. First, we take a particular care in preparing the electrolyte solutions by preliminary drying the salt, degassing the water solution and prepare the electrolyte in an argon-filled glovebox to be the most accurate possible about the salt concentration and storage condition. Then, we combine macroscopic viscosity and conductivity measurements with species-resolved diffusion from Pulsed-field gradient nuclear magnetic resonance (PFG-NMR) and dynamic nanostructure characterization via QENS. This multiscale approach allows us to capture both transient confinement and long-range dynamic, providing a unified view of transport in WISE, with and without an electrochemical stimulus.

2. Experimental part

2.1. Electrolyte preparation

A key challenge in preparing water-in-salt electrolytes is the precise control of water content. Lithium-ion battery salts, and LiTFSI in particular, are highly hygroscopic; if the salt is not properly dried before solution preparation, the final concentration may deviate from the targeted value. To minimize this uncertainty, LiTFSI was first dried overnight under vacuum at 120°C to remove residual water. The water was then degassed by argon bubbling for more than 20 min before being transferred into an Ar-filled glovebox. Finally, the dried salt was added according to the desired stoichiometry inside the glovebox, ensuring accurate control of the water content in the electrolyte. Because this protocol is not systematically applied in the literature, direct comparisons between reported concentrations should be made with caution.

All samples were prepared in a dedicated argon filled glove box with less than 1 ppm of O₂ and H₂O to ensure the best possible control of the salt concentration.

2.2. Viscosity measurement

Viscosity measurements were taken every 1°C between 50 and 5°C (except for 21 m and 18 m LiTFSI solution due to salt precipitation issue at low temperature) using a rolling-ball viscometer from Anton Paar. A liquid is introduced into a glass capillary (diameter 1.8 or 1.59 mm depending on the viscosity of the liquid) with a stainless-steel ball. The capillary must be filled with the liquid, and no gas bubble should be present in the capillary since it could alter the measurement. The capillaries were filled inside a glove box. Once the capillary is placed in the dedicated cavity, the capillary is tilted and the apparatus is measuring the speed of the ball, allowing the dynamic viscosity of the liquid to be determined.

2.3. Ionic conductivity measurement

The conductivity was measured using a Metrohm two-electrode type cell with platinum-plated electrodes. The cell is immersed in the probed solution and temperature is stabilized for at least 20 minutes before starting the measurement. The electrochemical impedance spectrum was recorded between 7 MHz and 1 Hz, with 5 mV perturbation amplitude, employing a VMP300 Biologic potentiostat.

2.4. NMR measurement

Self-diffusion coefficients of ⁷Li, ¹H and ¹⁹F between 25 and 45°C were measured by PFG-NMR using a Bruker 400 Avance III spectrometer (frequencies are 400.13, 376.50 and 155.51 MHz for ¹H, ¹⁹F and ⁷Li, respectively) with 5 mm diffusion probe and temperature regulation unit (maximum gradient 700 G.cm⁻¹). Self-diffusion coefficient of ¹H at 250 K and 273 K were measured by PFG-NMR using a Bruker

500 Avance II spectrometer (^1H frequency : 500.13 MHz) with a 5 mm BBI probe (maximum gradient 60 G.cm $^{-1}$). Temperature calibrations were performed with the standard chemical shifts method, and temperature accuracies and stabilities were better than ± 2 °C over the measurement time for each spectrometers. Standard pulsed field gradient stimulated echo sequences were used. Gradient pulse length δ was set at 1 ms for high temperature measurements and between 3 ms and 6 ms at low temperature, depending on the diffusion coefficient. Diffusion delay Δ were adjusted between 50 and 200 ms, and at extreme temperatures, at least two measurements were also performed with two different Δ values to confirm that no convection artifacts occur. Gradient strength was calibrated in measuring the self-diffusion of pure water. Repetition rate between scans was set to 5 T_1 (typically 10 to 20 s) to allow a complete relaxation of the spins and quantitative data. The self-diffusion coefficients were determined from the classical Stejskal-Tanner equation¹⁶.

2.5. QuasiElastic Neutron Scattering

QENS experiments were performed on SHARPER¹⁷ (run by the Laboratoire Léon Brillouin (LLB)) at Institut Laue-Langevin (ILL) (time-of-flight spectrometer (TOF), 70 μeV resolution)¹⁸, IN16B at ILL (backscattering spectrometer (BS), 0.7 μeV resolution)¹⁹, IN13 at ILL (backscattering spectrometer, 10 μeV resolution)²⁰, and FOCUS at Paul Scherrer Institute (PSI)²¹ (TOF, 40 μeV resolution). Samples were prepared in an Ar-filled glove box and sealed with indium gaskets in annular aluminum containers. ToF and BS acquired data were converted to $S(Q, \omega)$ spectra using the Mantid software²². The treated data (including empty container subtraction and normalization by vanadium) are then fitted with the home developed QENSH software²³. Resolution functions were measured at $\sim 2\text{K}$. Measurements were also performed applying a polarization of $\pm 1\text{V}$ in a modified QENS container with an electrolyte thickness of 150 μm (Note 3.6 - Figure S22).

3. Results & Discussion

3.1. Decoupling Between Viscosity and Conductivity

Viscosity and ionic conductivity of aqueous LiTFSI solutions were analyzed as a function of temperature and concentration using the Vogel–Fulcher–Tammann (VFT) model, appropriate for disordered and glass-forming systems (Figure 1, Figure 2). As expected, viscosity increases sharply with salt concentration, while ionic conductivity reaches a maximum around 5 mol.kg $^{-1}$ before declining slowly up to 10 mS.cm $^{-1}$ for 21 mol.kg $^{-1}$. The VTF model parameters are shown and discussed in **Note 1 (Supporting Information - Figure S1 and Figure S2)**. The results obtained are in good accordance with the ones previously published^{24,8, 25}. The conductivities and viscosities obtained for the WISE for 21 mol.kg $^{-1}$ LiTFSI are significantly higher than those obtained with a 1 mol.L $^{-1}$ LiTFSI in PC (Propylene

carbonate)²⁶, with values of $10 \text{ mS}\cdot\text{cm}^{-1}$ and $5 \text{ mS}\cdot\text{cm}^{-1}$ and $58 \text{ mPa}\cdot\text{s}$ and $7 \text{ mPa}\cdot\text{s}$, respectively, even though these two quantities typically exhibit opposite trends. To explain the high conductivity of WISEs despite their high viscosity, nanostructuration of WISE was proposed in literature.⁸

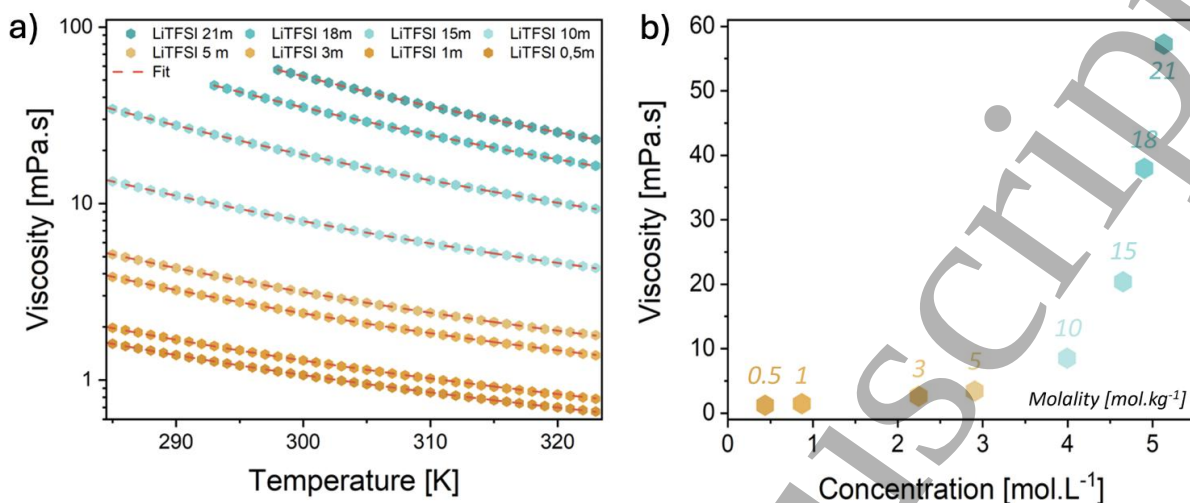


Figure 1 - Evolution of aqueous LiTFSI solution viscosity a) as a function of the temperature for various molality ($\text{mol}\cdot\text{kg}^{-1}$), b) measured at 25°C as a function of the concentration. The data were fitted with VTF model. Molalities (in $\text{mol}\cdot\text{kg}^{-1}$) of solutions are given as an indication directly in the figure on top of the symbols.

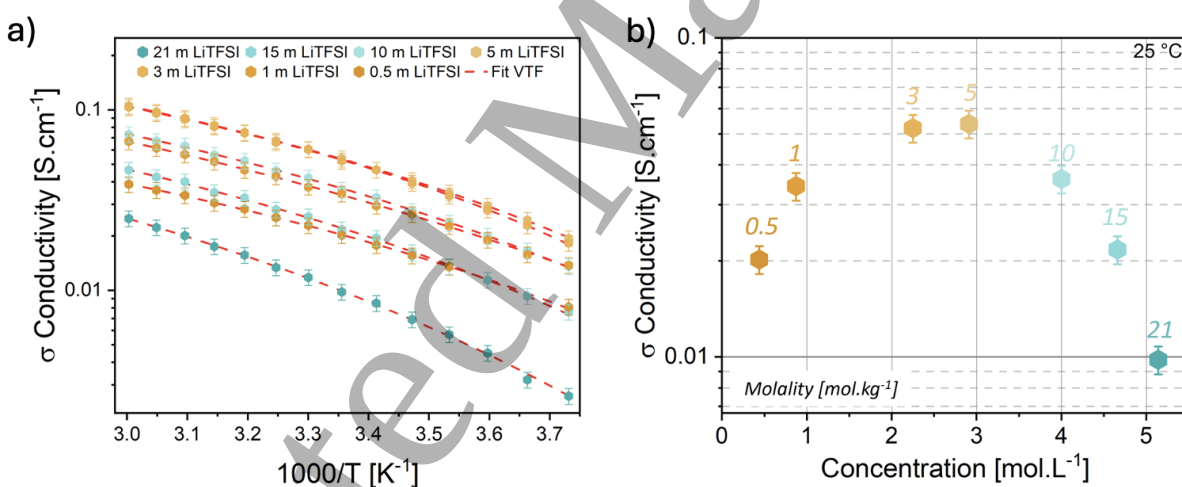


Figure 2 - Evolution of the ionic conductivity σ of aqueous LiTFSI solutions a) as a function of the temperature, b) at 25°C . The curves are fitted with a VTF model. Molalities (in $\text{mol}\cdot\text{kg}^{-1}$) of solutions are given as an indication directly in the figure on top of the symbols.

Walden's product remains constant as a function of the temperature only in the case of highly diluted solutions, because the salt is completely dissociated; this is no longer the case at high salt concentrations. To overcome this issue, the Walden equation was modified with the incorporation of a decoupling parameter γ , that can be calculated for each concentration²⁷⁻²⁹ Eq.1.

$$\Lambda \cdot \eta^\gamma = \text{cst} \quad \text{Eq.1}$$

with Λ ($\text{S.cm}^2.\text{mol}^{-1}$) the molar conductivity, η (mPa.s^{-1}) the viscosity and γ the decoupling parameter.

If the decoupling parameter is close to 1, the decoupling between ionic conductivity and viscosity is weak, meaning that the ionic conductivity and the viscosity are correlated. The salt is then fully dissociated, and the ionic conduction mechanism is the vehicular-type transport. The decoupling parameter in our case (**Figure 3**) decreases below unity with increasing salt concentration characterizing a decoupled transport. This behavior testifies a drift from classical hydrodynamic and has been studied in ionic liquid^{9,25}, where ion-ion correlations, steric hindrance, and local structuring lead to ionic motion via collective or hopping-like mechanisms that overcome viscous slow-down.

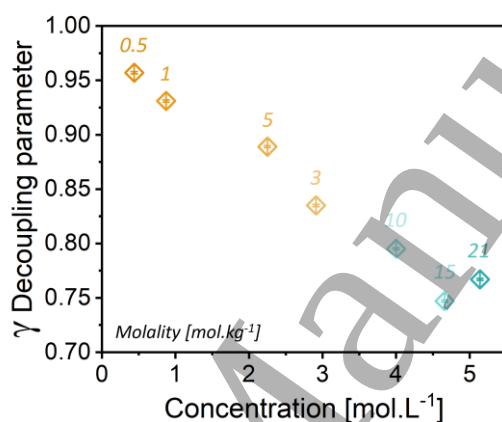


Figure 3 - Decoupling parameter γ evolution as a function of the salt concentration. The molality of the solution is given as an indication directly in the figure (number on top of the symbol).

3.3. Self-Diffusion Coefficients and Evolving Transport Mechanisms

To further probe the decoupling between ionic motion and bulk viscosity, we measured the self-diffusion coefficients of water (^1H), lithium ions (^7Li), and TFSI⁻ anions (^{19}F) using pulsed-field gradient NMR across a broad concentration range (0.5–21 mol.kg⁻¹) (**Figure 4**). The results are in good agreement with the one published at 25°C^{8,14}.

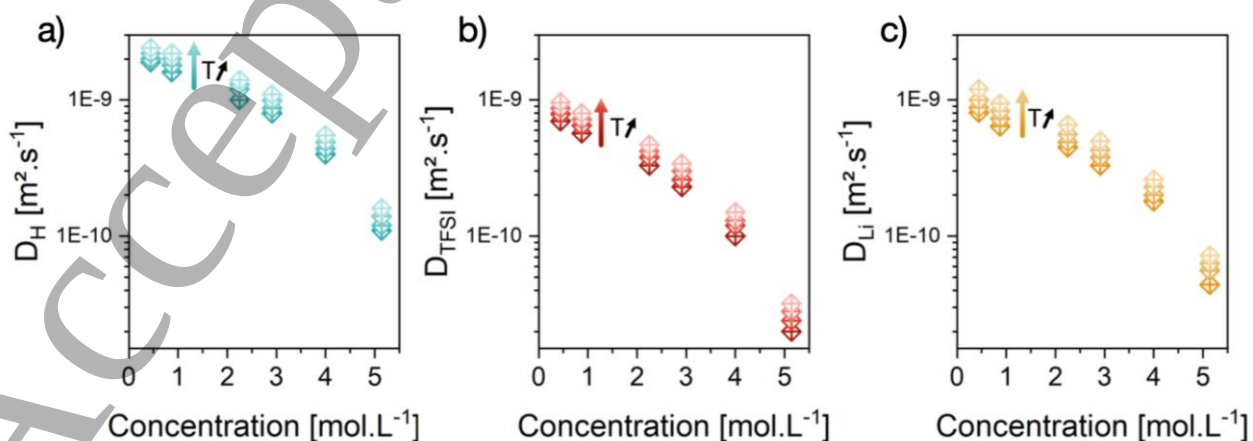


Figure 4 - Evolution of the self-diffusion coefficients of LiTFSI electrolyte obtained from PFG-NMR for a) D_{H_2O} b) D_{TFSI} and c) D_{Li} as function of the concentration. Temperature dependance is also highlighted on this figure.

At lower concentrations, self-diffusion coefficients of Li^+ and $TFSI^-$ are rather similar and small compared to thus of the water. As expected, all species exhibit decreasing mobility with increasing salt content due to rising viscosity. However, the extent of this slowdown is highly species dependent. Water and lithium species have self-diffusion coefficients that decrease by a factor of approximately 18 as the salt concentration increases from 0.5 to 21 mol·kg⁻¹, while TFSI species diffusion drops by more than a factor of 35. This disparity highlights that small and highly solvated cations retain significant mobility, whereas TFSI species becomes increasingly immobilized.

We further evaluated the species-specific decoupling parameter β (**Note 2 Supporting Information - Figure S3**), extracted from Stokes-Einstein relation. For TFSI species, β remains near 0.8 across all concentrations, indicating viscosity-governed motion. However, β for Li species and H_2O decrease with increasing salt content, confirming growing decoupling and selective ion mobility in structured environments (**Figure S3**). This ion-dependent transport behavior supports theoretical frameworks⁹, who showed that ionic crowding and asymmetry in nanoscale environments can yield emergent, species-specific pathways for transport. Recent MD simulations^{6,30} further revealed the formation of polymer-like Li^+ nanochains or percolating ionic networks in WISE, with short interionic distances stabilized by water and anions. These chains enable rapid lithium transport even in viscous environments, reinforcing the experimental observation of Lithium mobility persistence at high concentrations.

The apparent lithium transference number, determined by PFG-NMR (**Note 2 Supporting Information – Figure S4**) goes from approximately 0.54 for the lowest salt concentration, to 0.7 for the highest concentration, with a nearly linear trend for concentration higher than 1 mol·L⁻¹. Across the range of concentration investigated, the T_{app}^+ is independent from temperature between 25°C and 45°C. These high T_{app}^+ values and their increase with salt concentration have already been obtained by PFG-NMR³¹ and calculated by molecular dynamics (MD) simulations³², by neglecting the ion correlations. Nearly identical values were also obtained, at low salt concentration, using a Hittorf electrochemical cell, with a value of $T^+ \approx 0.5$ at 1 mol·L⁻¹.³³

However, for concentrated electrolytes, controversial values were obtained which might be related to the way the electrolytes are formulated and the technic used. With Hittorf measurement and MD simulation, considering the cation-anion correlations³², T_{Li}^+ decreases with salt concentration up to 0.3 and 0.22, at 5 mol·L⁻¹, respectively. However, the values obtained remain very high compared to those determined for organic electrolytes, considering cation-anion interactions, with as an example $T^+ =$

0.025 for LiTFSI dissolved in tetraglyme³⁴. The high value obtained for WISE is thus associated to the formation of an anionic network and the small mass of the $\text{Li}(\text{H}_2\text{O})_n^+$ species³².

By applying a constant current, instead of a potential like in the classic Bruce and Vincent method³⁵, a much lower Li^+ transference number, equal to 0.0075, was obtained at 5 mol.L^{-1} .³¹ However, from our experience, the 6 minutes measurement time is too short to reach a steady state, which could partly explain why the values are much lower than those reported using Hittorf measurement and MD simulation.

While total ionic conductivity can be easily determined, transference number determination is more difficult and several techniques were used such as electrochemical measurement (polarization method, Hittorf), PFG-NMR and electromotive force³⁶. In many cases, the techniques do not target the same species: PFG-NMR detects all species that self-diffuse, whereas electrochemical methods are mainly sensitive to charged species which move under an electric field. Furthermore, each method presents its own challenge. The Hittorf technique requires experimental conditions in which only migration transport occurs (no diffusion), whereas polarization method, the transference numbers obtained show notable inconsistencies in literature, mainly due to incorrect determination of the initial and steady-state currents, the establishment of the steady-state³⁷ or to the changes in cell impedance during the polarization measurement. Consequently, T^+ values depend on the methods used to determine it. However numerous experimental and modeling studies (NMR, Hittorf, MD) show rapid lithium transport consistent with the organization of concentrated electrolytes. In addition, the fast mobility of Li^+ ions in concentrated aqueous systems induces that no limitation due to mass transport have been reported during the operation of LFP/aqueous electrolyte battery under high charge and discharge C-rates³⁸.

3.4. The nanoscale origin of the decoupling between conductivity and viscosity

We performed quasi-elastic neutron scattering (QENS) measurements on highly concentrated LiTFSI solutions to investigate the dynamics of WISE at the molecular level. Due to the high incoherent scattering cross-section of hydrogen compared to all the other atoms, the QENS signal can be attributed to the H dynamics only. By combining time-of-flight (TOF) and backscattering (BS) spectrometers, we cover a broad range of characteristic times, probed from tenths of ps to few ns, providing a comprehensive view of the dynamics at the molecular level.

The QENS spectra were fitted using three Lorentzian components (**Figure 5**) allowing to identify three individual dynamical processes: i) the water self-diffusion (D_{SD} , $\text{nm}^2/\mu\text{s}$, ns – ms), ii) translational diffusion within confined environments (D_{loc} , $\text{\AA}^2/\text{ps}$, 1-10 ps), and iii) fast localized motion ($\text{\AA}$, 1 ps),

linked to molecular rotations. The total dynamical structure factor, $S_{\text{tot}}(Q, \omega)$, is the convolution of the individual contributions. It can be approximated as the sum of three Lorentzian of distinct widths (see **Note 3.4** for further details).

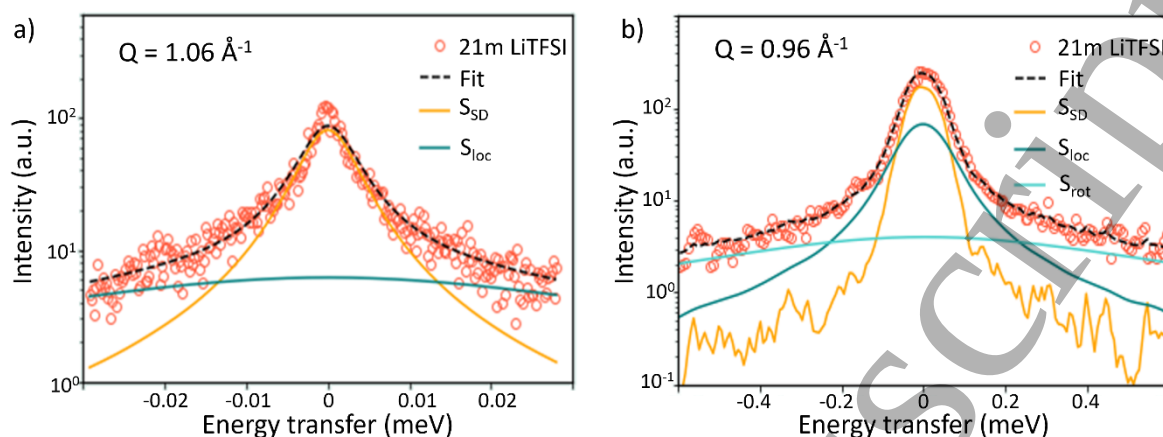


Figure 5 – Fitted dynamical structure factor $S(Q, \omega)$ of LiTFSI 21 m solution measured at two different energy resolutions on a) IN16B (BS) and b) FOCUS (TOF) at 303K. Both data sets are fitted with the same model involving three components: S_{SD} , S_{local} and S_{rot} (**Note 3**) corresponding to the water self-diffusion, translational localized diffusion and molecular rotations. Additional fits of both LiTFSI 21 m and 10 m are shown in **Figures S9, S10 and S13**.

The fitting procedure is fully detailed in the supporting information (**Note 3 - Figures S5-S15**) and follows a strategy employed to study the dynamics of ionic liquids charged with lithium salt and water confined in nanostructured ionic membranes. It relies on a systematic analysis (**Note 3**) to adjust two data sets covering a broad range of characteristic time (typically from the ps up to the ns). For each temperature and LiTFSI concentration, the 2 fitted data sets are cross-validated in terms of both width and integrals of each of the 3 Lorentzian components (**Figures S11-S12, S14-S15 and S18-S19** for both 10 m and 21 m). This meticulous procedure guarantees that the model proposed is in good agreement with both data sets.

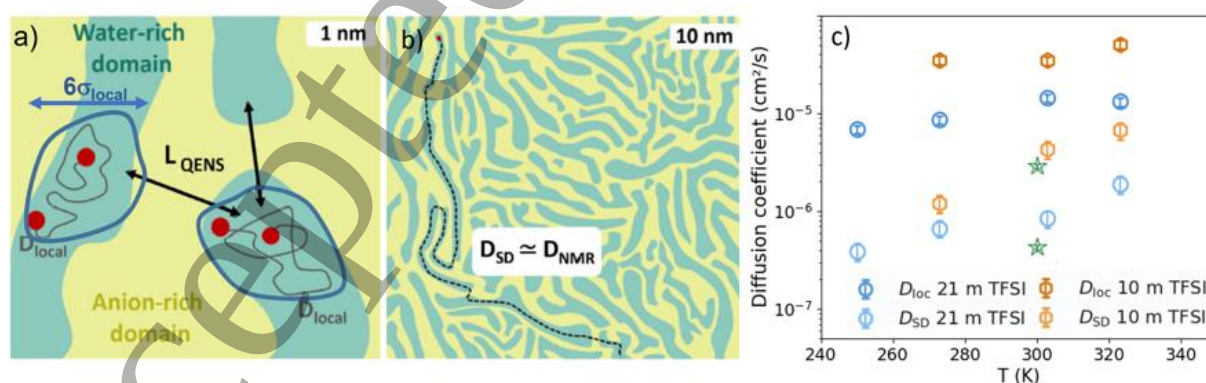


Figure 6 – a,b) Schematic representation of the transient nanostructure at two length scales together with the rapid localized diffusion within a domain and the self-diffusion at longer distance. c) Diffusion coefficients D_{local} and D_{SD} in LiTFSI 21 m and 10 m solutions. Data are compared with ref.¹⁴ (LiTFSI 21 m) in green.

Figure 6 shows a tentative representation of the transient nanostructure of WISE interpreted as water-rich nanodomains embedded in an anion-rich matrix that was first discovered by Borodin *et al.* using MD simulations and Small Angle Neutron Scattering (SANS).⁸ The associated dynamical parameters (D_{local} and D_{SD}), together with corresponding structural parameters (confinement sizes, $6\sigma_{\text{local}}$ and distance between domains, L_{QENS}) are shown in **Figure 6**, **Figure 7** and in **Figures S16-S17** and reported in **Tables S1-S2**.

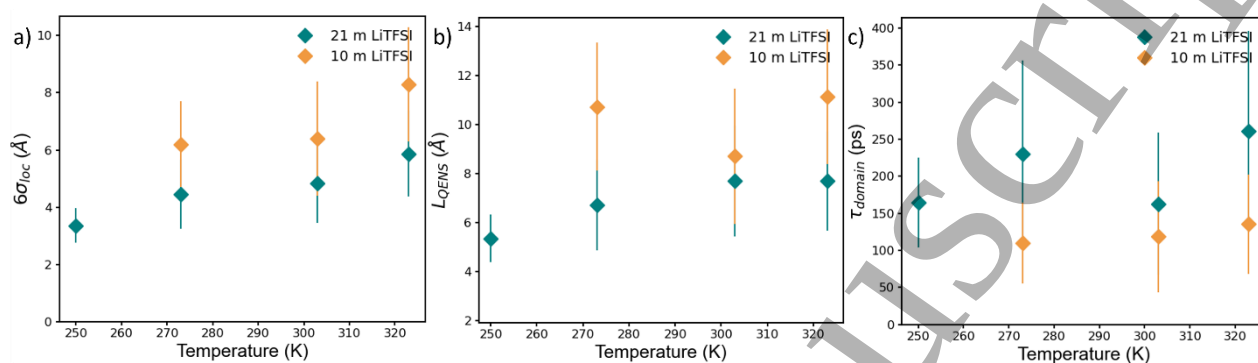


Figure 7 - Evolution of a) $6\sigma_{\text{local}}$, b) L_{QENS} and c) τ_{domain} as function of the temperature for 10 m and 21 m LiTFSI solutions.

Water molecules are found to show a fast translational diffusion (close to that of bulk water, **Figure 6**) confined in domains of few Å only (**Figure 7a**). At the nanoscale, water molecules can also diffuse to another confining domain (through the transient nanostructure), with a diffusion coefficient, D_{SD} , slower by a factor of ~ 10 (**Figure 6c**). The lower the temperature, the higher the ratio $D_{\text{local}} / D_{\text{SD}}$ is. Surprisingly, no additional slowdown of the dynamics is found at the micrometer scale (D_{NMR} measured by PFG-NMR $\sim D_{\text{SD}}$, **Table S1-S2**) revealing a rather weak nanostructuration that strengthen when the temperature decreases. Similar transient nanostructure has been shown in ionic liquids, where D_{NMR} can be up to an order of magnitude lower than D_{SD} .

Gonzalez *et al.*¹⁴ also extracted two diffusion coefficients from their analysis but obtained different values. They fitted their data using two Lorentzian components and a flat background, which led to a narrow component that appeared slower than the self-diffusion coefficient measured by PFG-NMR. They provided no convincing explanation for this discrepancy. The flat background term, while improving the statistical fit, does not correspond to a physical relaxation process and is replaced in our model by a third broader Lorentzian contribution, attributed to rotations of the water molecules. In contrast, the model developed here, explicitly separates the rotational/localised contributions (S_{rot} , S_{local}) from the long-range self-diffusion (S_{SD}), and resolves this inconsistency: $D_{\text{local}} > D_{\text{SD}}$ and D_{SD} is found to be in excellent agreement with D_{NMR} (**Figure S16 B**). This highlights the necessity of combining multiple experimental datasets when addressing multiscale dynamics. Regarding the distance between two water-rich domains, L_{QENS} , is of the order of 5 - 6 Å for the 21 m LiTFSI solution and around 1 nm for the 10 m LiTFSI one (**Figure 7b**). These values are in good agreement with previous studies realised

by Horwitz *et al.*³⁹, in which the distance between two domains was estimated to be around 1.2 nm for 21 m LiTFSI solution using SANS, and was found to increase at lower molality⁴⁰. The time needed to explore the confinement domain, τ_{domain} , is shown in **Figure 7c** as a function of temperature and is found to be (160 - 260) ps for the 21 m solution and slightly less for 10 m solution. This characteristic time could be attributed to the lifetime of a domain. Increasing the salt concentration induces a stronger nanostructuration with longer lived domains, and τ_{domain} is consistent with the relaxation time obtained by Liu *et al.* (~200 ps for 20 m LiTFSI) which reflects the reorganization of the ionic network⁴¹. We found the characteristic times of the reorientational and jump diffusion processes to be close to 1 ps and 8 ps, respectively, for LiTFSI 21m at 300K. While the reorientational process is similar to that of bulk water⁴², the residence time associated with water diffusion is approximately 8 times longer in 21m LiTFSI than in bulk water. Interestingly, both characteristic times are very close to the vibrational lifetimes of OD stretch modes of bulk-like and interfacial water, as measured by IR pump-probe spectroscopy⁴³. Although the interpretation of the data differs, Lim *et al.* also identified 2 dynamical processes within the ps time range: one close to bulk water (2 ps) and the other associated with interfacial water (~10 ps).

Borodin *et al.* were the first to propose the conceptual framework of interconnected, percolating Li⁺ water nanochannels enabling fast cation transport, on the basis of MD simulations and SANS.⁸ Altogether, the dynamic confinement model developed here, combined with PFG-NMR, conductivity and viscosity measurements gives additional insight into how small and highly solvated species such as lithium species can bypass viscous drag: they exploit long-lived but reconfigurable domains to form interconnected nanochannels and achieve efficient long-range transport. As concentration in lithium salt increases, the nanodomains persist longer, reinforcing the decoupling observed macroscopically between viscosity and self-diffusion. Our work provides dynamic experimental evidence for the structural model and add a temporal dimension that was previously inaccessible: we directly quantify the characteristic times associated with local self-diffusion within a domain and with inter-domain exchange. We also demonstrate how these timescales evolve with concentration and temperatures (**Figures S16-S17** and **Figure 7**). These results are also consistent with Faraone *et al.*¹⁵. By using QENS on a deuterated zinc-based WISE, they demonstrated that macroscopic viscosity scales with the lifetime of the anion-rich domains, while molar conductivity scales with the fluidity of the water/cation-rich domains. Faraone *et al.* used D₂O to highlight the coherent scattering to investigate collective modes and they analysed their data using a stretched-exponential model. This type of model is well suited to evaluating the distribution of relaxation time but does not allow to identify the physical nature of the motions (diffusion, rotation ...). The relaxation times of the anion domain of the Zinc based WISE at 20 m and 15m LiTFSI are found to be consistent with τ_{domain} : ~ 600 ± 400 ps and ~ 100 ± 50 ps, respectively.

Borodin et al. showed, using molecular dynamics simulations, that polarization in WISE can induce modifications of the cation solvation shell at electrode interfaces, leading to regions where water is either excluded or enriched⁴². Here, we investigate the impact of polarization on water dynamics using QENS under electrical polarization in a dedicated cell (**Figure S22 C**). The temperature dependence of the mean square displacement (MSD) and the dynamical structure factor of 21 m LiTFSI, measured with and without an applied polarization of 1 V (corresponding to an electric field of 67 V/cm), are shown in **Figure S22**. Both the MSD and the QENS spectra (recorded at 300 K) are nearly identical, indicating no detectable effect of polarization on water dynamics: at the ps – ns timescale the dynamics probed by QENS is insensitive to the application of couple of volts into the electrolyte. This result is consistent with Martinez et al.⁴⁴ that performed *operando* QENS in a fuel cell (i.e. they probe water dynamics in a nanoporous and charged matrix). Modification of translational diffusion has been reported for much higher electric field (2.5 kV/mm)⁴⁵. It should be noted that QENS probes an average over both interfacial and bulk contributions. In the present geometry, the interfacial region (a few nanometers) is negligible compared to the bulk electrolyte thickness (~150 μm) and therefore remains below the detection limit. These results suggest that the dynamical parameters extracted from equilibrium QENS measurements can be reasonably transposed to fully *operando* electrochemical cells.

4. Conclusion

The combined insights from conductivity, PFG-NMR and QENS measurements converge on a coherent picture of hierarchical transport in WISE. At low salt concentrations, transport behavior aligns with classical hydrodynamic models: viscosity and conductivity scale together.

However, as the concentration increases, this coupling progressively breaks down. Lithium species and protons retain substantial mobility, while TFSI⁻ becomes increasingly hindered. This species-specific decoupling points to a shift from purely vehicular motion to a mixed mechanism: localized, rapid hopping within transient water-rich nanodomains, combined with interdomain exchange as represented in **Figure 6**.

Our multiscale experimental approach directly captures this transition. QENS reveals confined, sub-nanosecond dynamics within transient domains, while PFG-NMR tracks longer-range, species-dependent self-diffusion. The resulting transport framework is both spatially and temporally heterogeneous. Fast, localized self-diffusion occurs within transient water-rich domain ($\tau \sim 1\text{--}20$ ps). Longer-range motion proceeds via dynamic exchange between domains ($\tau \sim 100$ ps). This study highlights how transient nanostructuration can sustain high mobility in viscous media. By spanning multiple time and length scales, our work provides direct experimental evidence for a hierarchical

transport model, offering a new lens/framework through which to interpret and design next generation superconcentrated electrolytes. Moreover, it delivers key dynamical parameters that directly feed and constrain molecular dynamics and machine-learning-assisted molecular dynamics simulations.

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Supporting information

Supporting information give details on conductivity and viscosity measurements (Note 1, Figures S1-S2), self-diffusion coefficient, decoupling factor and apparent transference numbers (Note 2, Figure S3-S4) and QENS (Note 3). Note 3 contains the methodology for the QENS fitting and analysis (Note 3.1, Figures S5-S8), results on 21 m LiTFSI (Note 3.2, Figures S9-S12, Table S1), results on 10 m LiTFSI (Note 3.3, Figures S13-S17, Table S2), the dynamical structure factor modeling (Note 3.4, Figures S18-S19), dynamical confinement sizes (Note 3.5, Figures S20-S21), and QENS under electrical polarization (Note 3.6, Figure S22).

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